# HASTS 416 Tutorial

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- Define the Transition Probability Matrix
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- Define the Transition Probability Matrix (7 × 7)
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```
# Function to save and display plots
save_and_show <- function(plot, filename, width = 8, height = 5) {
  # Save the plot
  ggsave(filename, plot, width = width, height = height, dpi = 100)
  # Display in markdown
  knitr::include_graphics(filename)
  # Return plot invisibly
  invisible(plot)
}
```

#Question A1

## 5-State Markov Chain Analysis

```
# Load required libraries
library(markovchain)
```

```
## Loading required package: Matrix

## Package:  markovchain
```

```
## Version:  0.10.3
## Date:     2026-02-02 06:30:37 UTC
## BugReport: https://github.com/spedygiorgio/markovchain/issues
```

```
library(diagram)
```

```
## Loading required package: shape
```

```
library(ggplot2)
```

```
library(reshape2)
library(expm)
```

```
##
## Attaching package: 'expm'

## The following object is masked from 'package:Matrix':
##
##     expm
```

# Define the Transition Probability Matrix

```
P <- matrix(c(
  0.1, 0.3, 0.2, 0.2, 0.2,
  0.2, 0.1, 0.3, 0.2, 0.2,
  0.0, 0.0, 0.5, 0.5, 0.0,
  0.0, 0.0, 0.5, 0.4, 0.1,
  0.0, 0.0, 0.0, 0.0, 1.0
), nrow = 5, byrow = TRUE)

rowSums(P)
```

```
## [1] 1 1 1 1 1
```

```
colnames(P) <- 1:5
rownames(P) <- 1:5
P
```

```
##     1   2   3   4   5
## 1 0.1 0.3 0.2 0.2 0.2
## 2 0.2 0.1 0.3 0.2 0.2
## 3 0.0 0.0 0.5 0.5 0.0
## 4 0.0 0.0 0.5 0.4 0.1
## 5 0.0 0.0 0.0 0.0 1.0
```

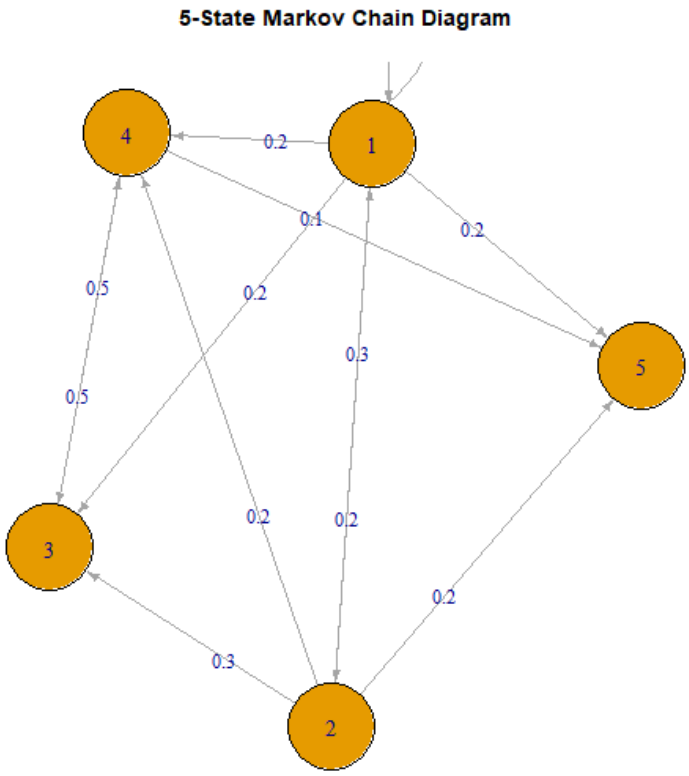
# (a) Markov Chain Classification

- Plot diagram
- Identify classes
- Determine periods
- Identify absorbing & reflective states

```
# Create Markov chain object
mc <- new("markovchain",
          states = as.character(1:5),
          transitionMatrix = P)

# Plot and save - METHOD 1: Using plot() directly
png("A1_markov_diagram.png", width = 800, height = 600)
plot(mc,
     main = "5-State Markov Chain Diagram",
     edge.arrow.size = 0.5,
     vertex.size = 30,
     vertex.label.cex = 1.2)
dev.off()
```

```
## png
##   2
```

```
# Display the saved plot
knitr::include_graphics("A1_markov_diagram.png")
```




```
# Summary
summary(mc)
```

```
## Unnamed Markov chain  Markov chain that is composed by:
## Closed classes:
## 5
## Recurrent classes:
## {5}
## Transient classes:
## {1,2},{3,4}
## The Markov chain is not irreducible
## The absorbing states are: 5
```

```
# Communicating classes
classes <- communicatingClasses(mc)
classes
```

```
## [[1]]
## [1] "1" "2"
##
## [[2]]
## [1] "3" "4"
```

```
##
## [[3]]
## [1] "5"
```

```
# Class type (recurrent/transient)
class_types <- character(length(classes))

for(i in 1:length(classes)){
  class_states <- classes[[i]]
  is_closed <- TRUE

  for(state in class_states){
    for(next_state in 1:5){
      if(P[as.numeric(state), next_state] > 0 &&
         !(as.character(next_state) %in% class_states)){
        is_closed <- FALSE
        break
      }
    }
  }

  class_types[i] <- ifelse(is_closed, "RECURRENT", "TRANSIENT")
}

class_types
```

```
## [1] "TRANSIENT" "TRANSIENT" "RECURRENT"
```

```
#Periods
period(mc)
```

```
## Warning in period(mc): The matrix is not irreducible

## [1] 0
```

```
# Absorbing states
absorbing_states <- which(diag(P) == 1)
absorbing_states
```

```
## 5
## 5
```

```
# Reflective states (no self-loop)
reflective_states <- which(diag(P) == 0)
reflective_states
```

```
## named integer(0)
```

# (b) Simulate Three Trajectories

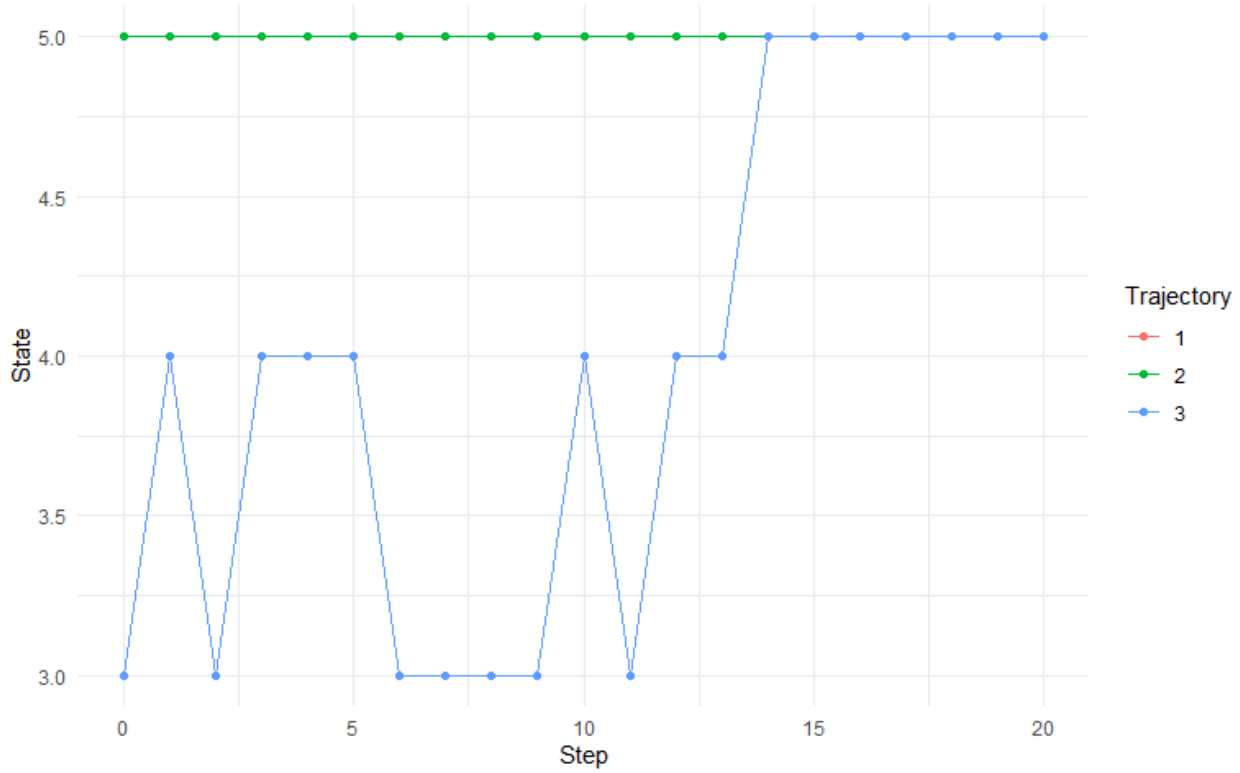
```
set.seed(123)

simulate_trajectory <- function(start_state, n_steps = 20){
  traj <- numeric(n_steps + 1)
  traj[1] <- start_state

  for(i in 1:n_steps){
    start_state <- sample(1:5, 1, prob = P[start_state, ])
    traj[i + 1] <- start_state
  }

  traj
}
```

```
set.seed(456)
start_states <- sample(1:5, 3, replace = TRUE)
start_states
```

```
## [1] 5 5 3
```

```
trajectories <- lapply(start_states, simulate_trajectory)
```

```
# Plot trajectories
traj_data <- data.frame(
  Step = rep(0:20, 3),
  State = unlist(trajectories),
  Trajectory = factor(rep(1:3, each = 21))
)

ggplot(traj_data, aes(Step, State, color = Trajectory)) +
  geom_line() +
  geom_point() +
  theme_minimal()
```




## (c) Steady-State Probabilities

```
# Check if chain is irreducible
is_irreducible <- is.irreducible(mc)

# Check if chain is aperiodic (period = 1 for whole chain)
is_aperiodic <- (period(mc) == 1)
```

```
## Warning in period(mc): The matrix is not irreducible
```

```
# Ergodic if irreducible and aperiodic
is_ergodic <- is_irreducible && is_aperiodic
is_ergodic
```

```
## [1] FALSE
```

```
if(is_ergodic){
  steadyStates(mc)
}
```

```
# Solve manually
A <- t(P) - diag(5)
A <- rbind(A, rep(1, 5))
b <- c(rep(0, 5), 1)
```

```
steady <- solve(t(A) %*% A, t(A) %*% b)
steady / sum(steady)
```

```
##   [,1]
## 1    0
## 2    0
## 3    0
## 4    0
## 5    1
```

```
# Solve manually
A <- t(P) - diag(5)
A <- rbind(A, rep(1, 5))
b <- c(rep(0, 5), 1)

steady <- solve(t(A) %*% A, t(A) %*% b)
steady / sum(steady)
```

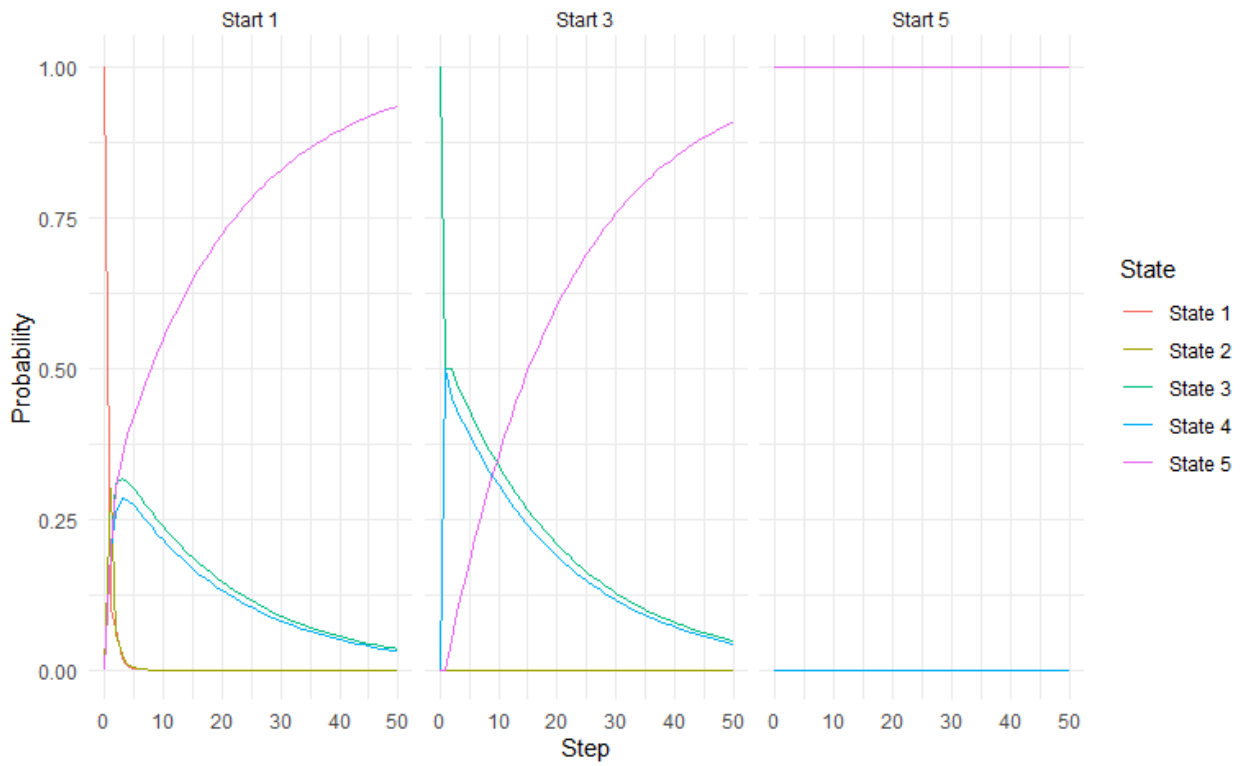
```
##   [,1]
## 1    0
## 2    0
## 3    0
## 4    0
## 5    1
```

# (d) Convergence to Steady-State

```
n_steps <- 50
start_states_conv <- c(1, 3, 5)

prob_over_time <- array(0, dim = c(3, n_steps + 1, 5))
```

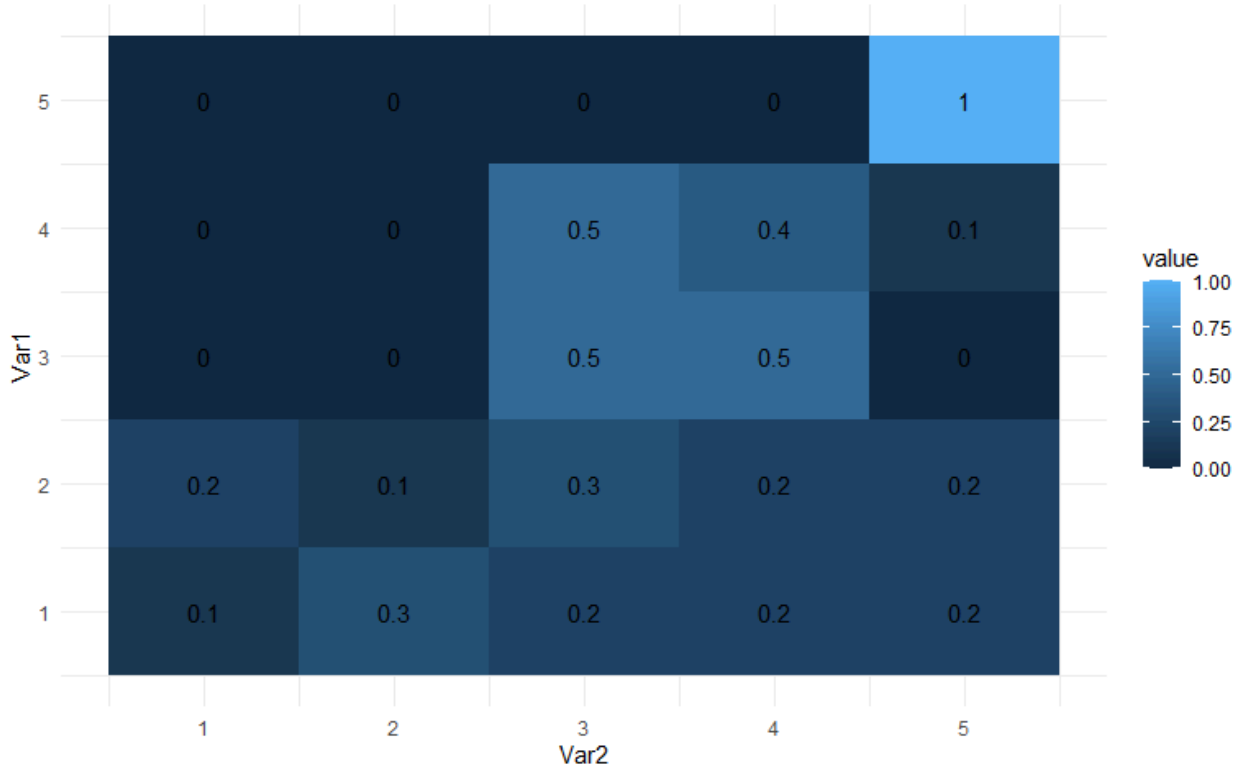
```
for(i in 1:3){
  dist <- rep(0, 5)
  dist[start_states_conv[i]] <- 1
  prob_over_time[i, 1, ] <- dist

  for(step in 2:(n_steps + 1)){
    dist <- dist %*% P
    prob_over_time[i, step, ] <- dist
  }
}
```

```
# Plot convergence
plot_data <- data.frame()

for(i in 1:3){
  for(s in 1:5){
    plot_data <- rbind(plot_data, data.frame(
      Step = 0:n_steps,
      Probability = prob_over_time[i,,s],
      Start = paste("Start", start_states_conv[i]),
      State = paste("State", s)
    ))
  }
}

ggplot(plot_data, aes(Step, Probability, color = State)) +
  geom_line() +
  facet_wrap(~Start) +
  theme_minimal()
```




## Transition Matrix Heatmap

```
P_melted <- melt(P)

ggplot(P_melted, aes(Var2, Var1, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value,2))) +
  theme_minimal()
```




#Question A2

## 7-State Markov Chain Analysis

# Define the Transition Probability Matrix (7 × 7)

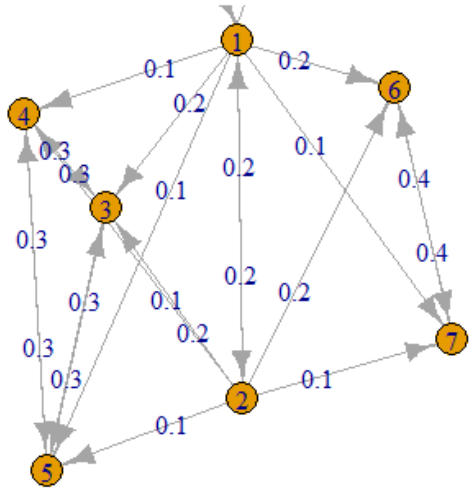
```
P <- matrix(c(
  0.1, 0.2, 0.2, 0.1, 0.1, 0.2, 0.1,
  0.2, 0.1, 0.2, 0.1, 0.1, 0.2, 0.1,
  0.0, 0.0, 0.4, 0.3, 0.3, 0.0, 0.0,
  0.0, 0.0, 0.3, 0.4, 0.3, 0.0, 0.0,
  0.0, 0.0, 0.3, 0.3, 0.4, 0.0, 0.0,
  0.0, 0.0, 0.0, 0.0, 0.0, 0.6, 0.4,
  0.0, 0.0, 0.0, 0.0, 0.0, 0.4, 0.6
), nrow = 7, byrow = TRUE)

round(rowSums(P), 2)
```

```
## [1] 1 1 1 1 1 1 1
```

# (a) Plot the Markov Chain Diagram

```
mc <- new("markovchain",
          states = as.character(1:7),
          transitionMatrix = P)
```

```
plot(mc)
```




```
# Save diagram
#dev.copy(png, "A2_diagram.png", width = 800, height = 600)
#dev.off()
```

# (b) Classification of States

#Communicating Classes

```
classes <- communicatingClasses(mc)
classes
```

```
## [[1]]
## [1] "1" "2"
##
## [[2]]
## [1] "3" "4" "5"
##
## [[3]]
## [1] "6" "7"
```

#Transient Vs Recurrent

```
class_types <- character(length(classes))

for(i in 1:length(classes)){
  class_states <- classes[[i]]
  is_closed <- TRUE

  for(state in class_states){
    s_idx <- as.numeric(state)
    for(next_s in 1:7){
      if(P[s_idx, next_s] > 0 &&
         !(as.character(next_s) %in% class_states)){
        is_closed <- FALSE
        break
      }
    }
  }

  class_types[i] <- ifelse(is_closed, "RECURRENT", "TRANSIENT")
}

class_types
```

```
## [1] "TRANSIENT" "RECURRENT" "RECURRENT"
```

#Period

```
period(mc)
```

```
## Warning in period(mc): The matrix is not irreducible

## [1] 0
```

#Absorbing states

```
absorbing <- which(diag(P) == 1)
absorbing
```

```
## integer(0)
```

#Reflecting states

```
reflecting <- which(diag(P) == 0)
```

```
reflecting


## integer(0)
```

# Transition Matrix Heatmap

```
P_melt <- melt(P)
colnames(P_melt) <- c("From", "To", "Prob")

ggplot(P_melt, aes(x = factor(To), y = factor(From), fill = Prob)) +
  geom_tile() +
  geom_text(aes(label = round(Prob, 2))) +
  scale_fill_gradient(low = "white", high = "steelblue") +
  theme_minimal()
```

```
ggsave("A2_heatmap.png", width = 8, height = 6)
```

# Question A3

# Time-Inhomogeneous Markov Chain (Traffic Conditions)

# Define State Space and Transition Matrices

```
states <- c("Light", "Heavy", "Jammed")

P1 <- matrix(c(
  0.4, 0.4, 0.2,
  0.3, 0.4, 0.3,
  0.0, 0.1, 0.9
), nrow = 3, byrow = TRUE)

P2 <- matrix(c(
  0.1, 0.5, 0.4,
  0.1, 0.3, 0.6,
  0.0, 0.1, 0.9
), nrow = 3, byrow = TRUE)

round(rowSums(P1), 2)
```

```
## [1] 1 1 1
```

```
round(rowSums(P2), 2)
```

```
## [1] 1 1 1
```

# Transition Matrices

```
colnames(P1) <- states; rownames(P1) <- states
colnames(P2) <- states; rownames(P2) <- states

P1
```

```
##        Light Heavy Jammed
## Light    0.4   0.4    0.2
## Heavy    0.3   0.4    0.3
## Jammed   0.0   0.1    0.9
```

```
P2
```

```
##        Light Heavy Jammed
## Light    0.1   0.5    0.4
## Heavy    0.1   0.3    0.6
## Jammed   0.0   0.1    0.9
```

# (a) Theoretical Distribution at 6PM

```
initial <- c(1, 0, 0)

# 4PM
dist_4PM <- initial
for(i in 1:9){
  dist_4PM <- dist_4PM %*% P1
}
dist_4PM
```

```
##          Light     Heavy    Jammed
## [1,] 0.1005492 0.1899846 0.7094662
```

```
# 6PM
dist_6PM <- dist_4PM
for(i in 1:6){
  dist_6PM <- dist_6PM %*% P2
}
dist_6PM
```

```
##           Light    Heavy    Jammed
## [1,] 0.01480077 0.132596 0.8526032
```

```
# Verification
dist_6PM_alt <- initial %*% (P1 %^% 9 %*% P2 %^% 6)
dist_6PM_alt
```

```
##           Light     Heavy    Jammed
## [1,] 0.02710305 0.1722042 0.8006928
```

# (b) Simulation of 10,000 Trajectories

```
set.seed(123)

simulate <- function(){
  state <- 1

  for(i in 1:9){
    state <- sample(1:3, 1, prob = P1[state, ])
  }

  for(i in 1:6){
    state <- sample(1:3, 1, prob = P2[state, ])
  }

  state
}


n_sim <- 10000
results <- replicate(n_sim, simulate())

empirical <- table(results) / n_sim
names(empirical) <- states

empirical
```

```
##  Light  Heavy Jammed
## 0.0146 0.1287 0.8567
```

# Theoretical vs Empirical Comparison

```
comparison <- data.frame(
  State = states,
  Theoretical = dist_6PM,
  Empirical = as.numeric(empirical),
  Difference = dist_6PM - as.numeric(empirical)
)

comparison
```

```
##    State Theoretical.Light Theoretical.Heavy Theoretical.Jammed Empirical
## 1  Light        0.01480077          0.132596          0.8526032    0.0146
## 2  Heavy        0.01480077          0.132596          0.8526032    0.1287
```

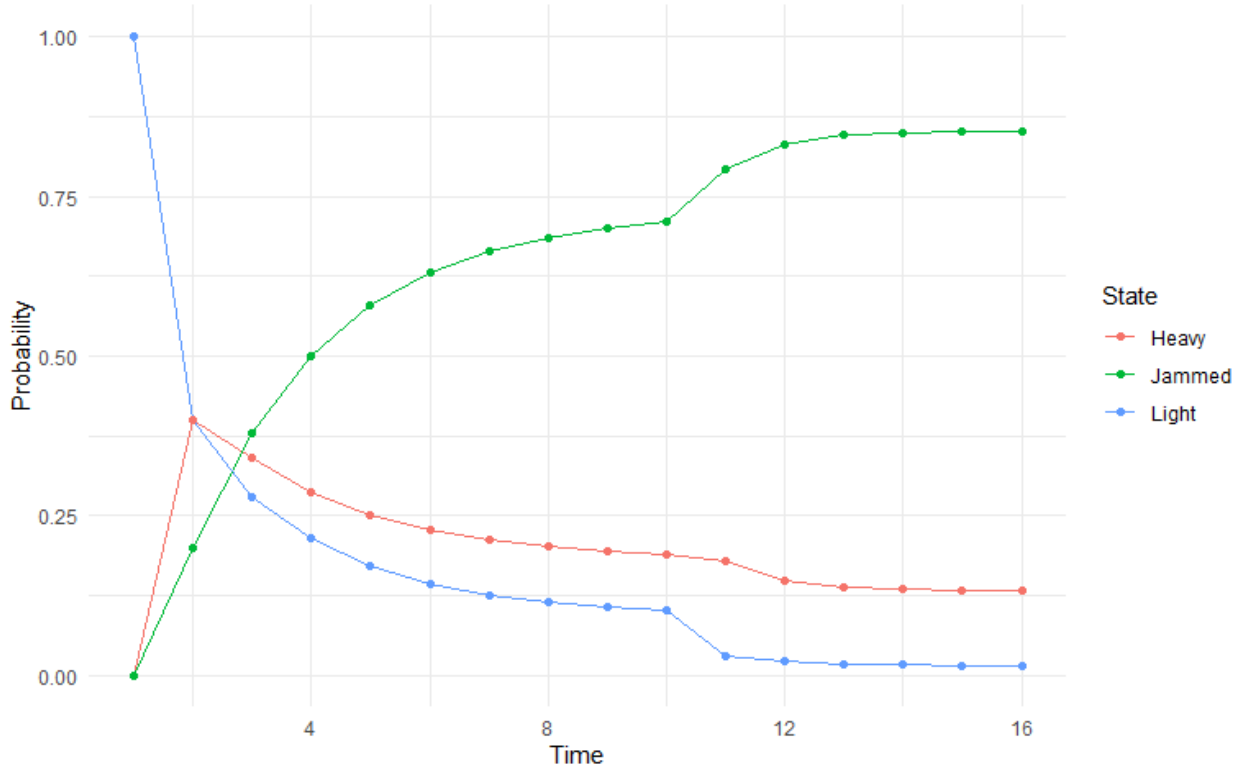
```
## 3 Jammed        0.01480077          0.132596          0.8526032    0.8567
##   Difference.Light Difference.Heavy Difference.Jammed
## 1     0.0002007669      0.003895988      -0.004096755
## 2     0.0002007669      0.003895988      -0.004096755
## 3     0.0002007669      0.003895988      -0.004096755
```

```
chisq.test(x = as.numeric(empirical) * n_sim, p = dist_6PM)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  as.numeric(empirical) * n_sim
## X-squared = 1.3688, df = 2, p-value = 0.5044
```

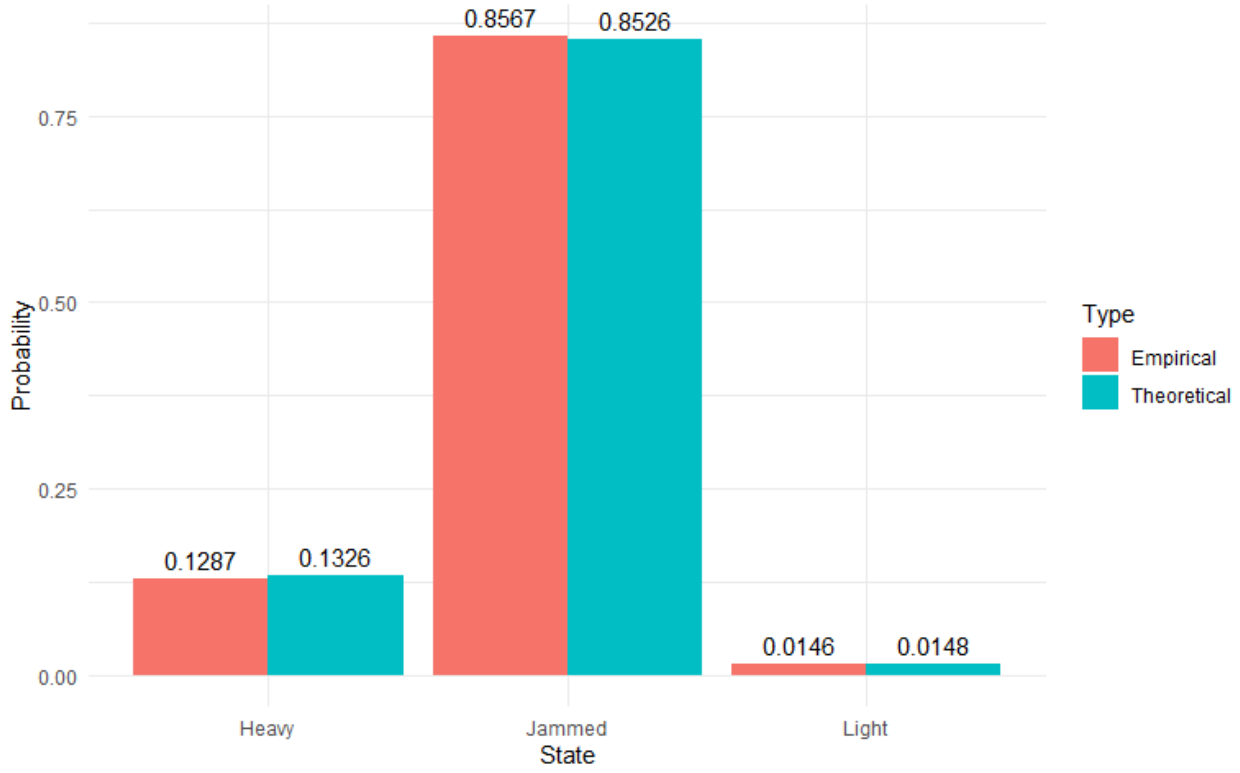
# Evolution of Probabilities Over Time

```
time_labels <- c("1PM","1:20","1:40","2PM","2:20","2:40",
                 "3PM","3:20","3:40","4PM","4:20","4:40",
                 "5PM","5:20","5:40","6PM")

dist_time <- matrix(0, nrow = 16, ncol = 3)
dist_time[1, ] <- initial

for(t in 2:10){
  dist_time[t, ] <- dist_time[t-1, ] %*% P1
}
for(t in 11:16){
  dist_time[t, ] <- dist_time[t-1, ] %*% P2
}

plot_data <- data.frame(
  Time = rep(1:16, 3),
  State = rep(states, each = 16),
  Probability = as.vector(dist_time)
)

ggplot(plot_data, aes(Time, Probability, color = State)) +
  geom_line() +
  geom_point() +
  theme_minimal()
```



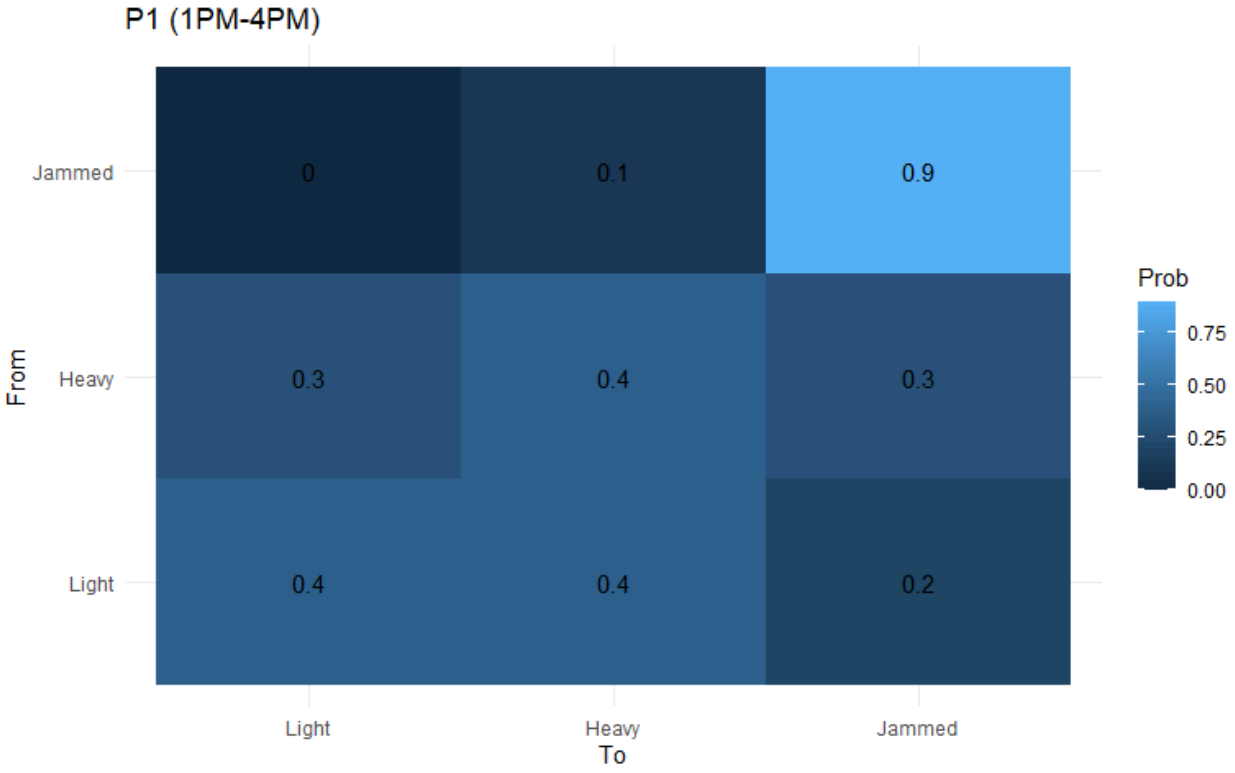

## Final Distribution Comparison

```
comp_data <- data.frame(
  State = rep(states, 2),
  Type = rep(c("Theoretical", "Empirical"), each = 3),
  Probability = c(dist_6PM, as.numeric(empirical))
)

ggplot(comp_data, aes(State, Probability, fill = Type)) +
  geom_bar(stat = "identity", position = "dodge") +
  geom_text(aes(label = round(Probability, 4)),
            position = position_dodge(0.9), vjust = -0.5) +
  theme_minimal()
```

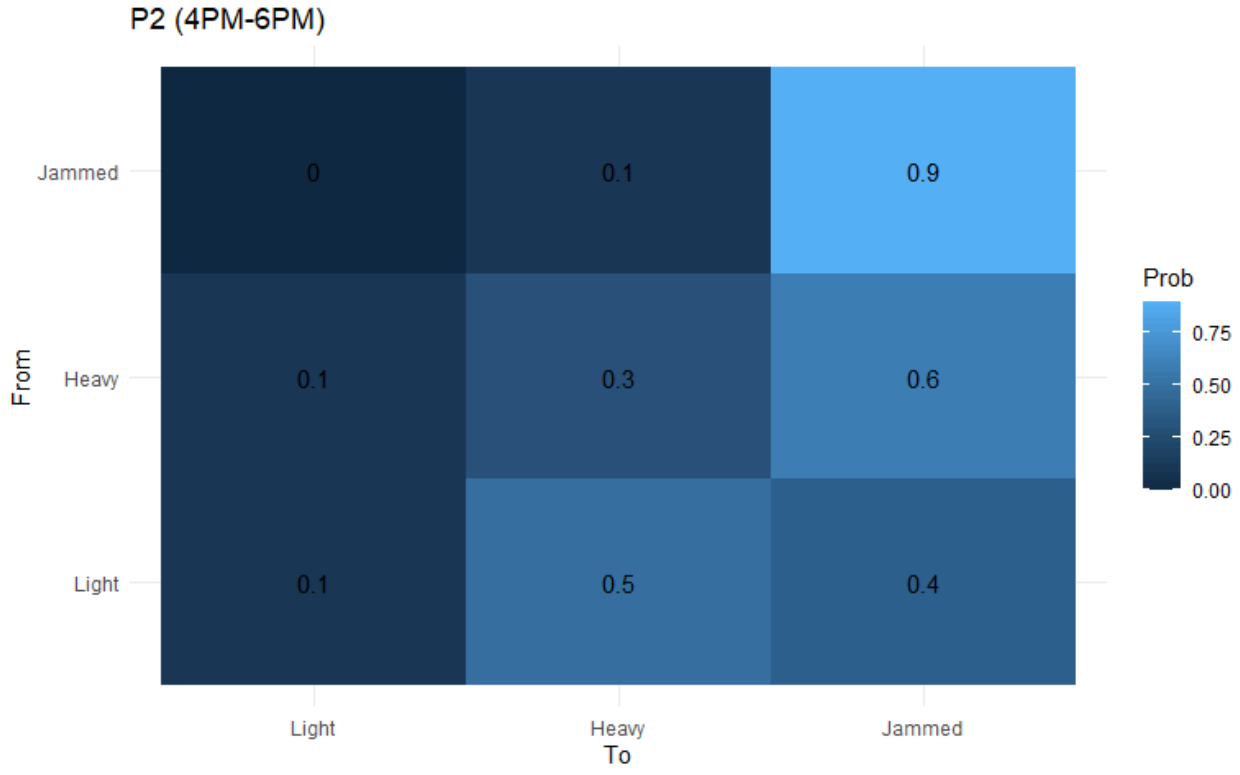



## Transition Matrix Heatmaps

```
# Heatmap for P1
P1_melt <- melt(P1)
colnames(P1_melt) <- c("From","To","Prob")

ggplot(P1_melt, aes(To, From, fill = Prob)) +
  geom_tile() +
  geom_text(aes(label = round(Prob,2))) +
  ggtitle("P1 (1PM-4PM)") +
  theme_minimal()
```




```
# Heatmap for P2
P2_melt <- melt(P2)
colnames(P2_melt) <- c("From","To","Prob")

ggplot(P2_melt, aes(To, From, fill = Prob)) +
  geom_tile() +
  geom_text(aes(label = round(Prob,2))) +
  ggtitle("P2 (4PM-6PM)") +
  theme_minimal()
```


P2 (4PM-6PM)
Jammed
Heavy
Light
From
0
0.1
0.9
0.1
0.3
0.6
0.1
0.5
0.4
Light
Heavy
Jammed
To
Prob
0.75
0.50
0.25
0.00